Combos, Not Spam: A Human-Like Super Smash Bros. Melee Bot

Your Name, Teammate Name

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Motivation

Can a bot learn to combo instead of spam?

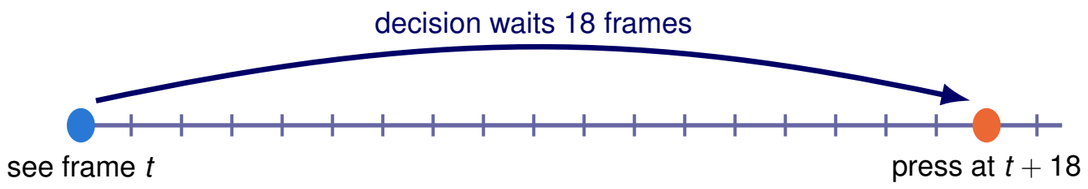


Figure 1: Our bot (left) ends a forward-air → drill → shine combo on the level 9 CPU: Fox's signature combo, captured from the emulator.

- **Melee**: 60 frames/s, frame-precise technique. Deep RL has beaten pros [5], but our first PPO bot won by **spamming one move**.
- **Goal**: a Fox that is strong *and* plays like a human.

Problem formulation

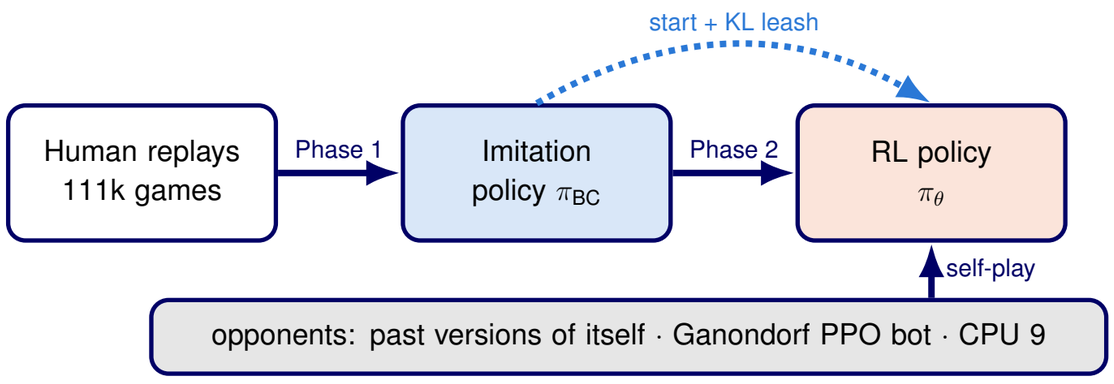
- **Observe** h_t: both players' position, speed, damage, action state, …
- **Act** a_t: a full controller per frame, $48 \cdot 17 \cdot 17 \cdot 5 \approx 69$k choices (buttons, stick x, y, C-stick).
- **React like a human** [4]: 18 frames (300 ms) late.



Objective: win, but stay close to how humans play:

$$\max_\theta \; \mathbb{E}_{\pi_\theta}\Big[\sum_t \gamma^t r_t\Big] - \beta \sum_t \mathrm{KL}\big(\pi_\theta(\cdot \mid h_t) \,\big\|\, \pi_{\text{human}}(\cdot \mid h_t)\big)$$

Method: two phases



Phase 1: learning from humans

- **Data**: 111,310 ranked games of master and diamond Fox players [2], 1.24 billion frames.
- **Policy**: 3 × (residual LSTM [6] + MLP), width 512, autoregressive heads; 9.9 M parameters, as in slippi-ai [3].

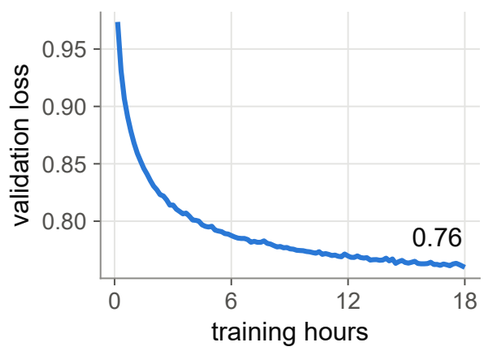


Figure 2: 18 h on one laptop GPU. Accuracy: buttons 93%, sticks 89/94%, C-stick 99%.

Behaviour cloning: predict the human's controller 18 frames ahead:

$$\mathcal{L}_{\text{BC}}(\theta) = -\sum_t \sum_{k=1}^{4} \log \pi_\theta\big(a_{t+18}^{(k)} \,\big|\, h_t,\; a_{t+18}^{(<k)}\big)$$

Phase 2: reinforcement learning on a leash

PPO [9] + a KL penalty to the frozen imitation policy [7, 10]:

$$\mathcal{L}(\theta) = \mathbb{E}_t\Big[-\min\big(\rho_t \hat{A}_t,\; \mathrm{clip}(\rho_t, 1 \pm \epsilon)\,\hat{A}_t\big) + \beta\,\mathrm{KL}\big(\pi_\theta \,\|\, \pi_{\text{BC}}\big)(h_t)\Big]$$

$$r_t = \Delta\text{stocks}_{\text{opp}} - \Delta\text{stocks}_{\text{own}} + 0.01\,\big(\Delta\%_{\text{opp}} - \Delta\%_{\text{own}}\big) + r_t^{\text{shape}}$$

- r_t^{shape}: small anti-exploit terms; GAE [8], $\gamma = 0.997$, $\beta = 0.5$, $\epsilon = 0.2$.
- 10 parallel games (libmelee [1]): 6 self-play, 2 Ganondorf, 2 CPU 9.

Results: win rates after 3 h of RL

100% vs CPU level 9 — imitation: 95%

95% vs Ganondorf bot — imitation: 65%

62% vs imitation — its own teacher

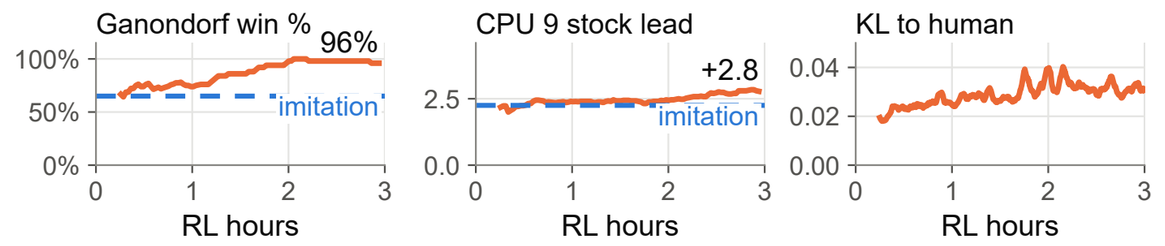
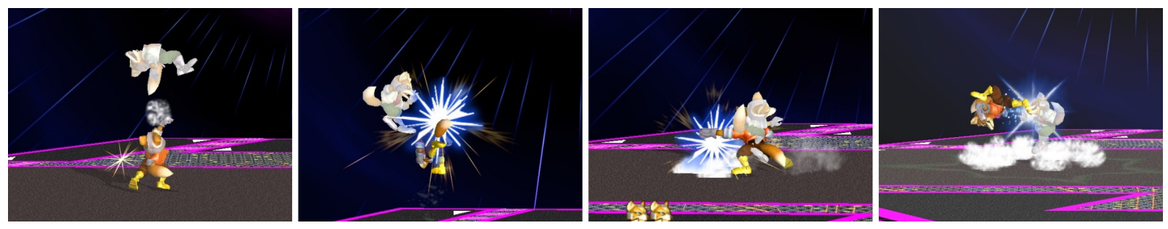


Figure 3: During RL: rolling mean over 50 games (dashed: imitation). The distance to the human policy levels off at 0.03 nats/frame. Final Destination, 4 stocks, 20–40 games per test.

Results: still human-like



1 up-throw 127% **2** up-air 138% **3** jab 141% **4** up-smash 154%

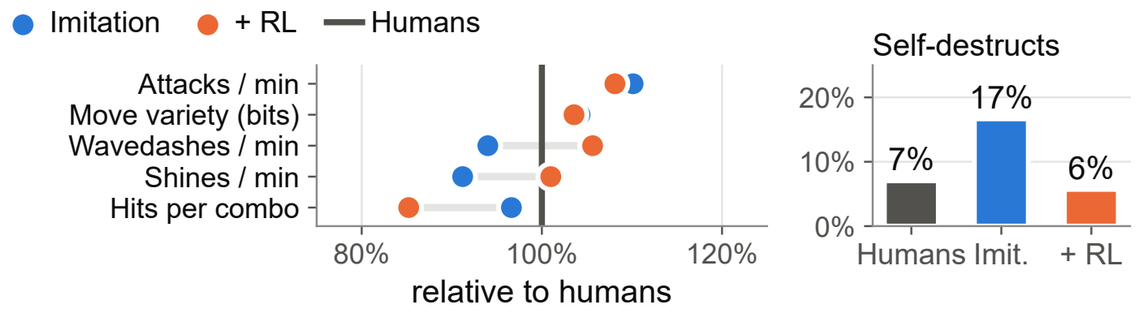


Figure 4: Top: a combo by our bot, frame by frame, with the opponent's damage. Bottom: play style against 22 ranked human Fox players; self-destruct: a death without being hit in the 4 s before.

Self-destructs 17% → **6%** (humans 7%); combos a bit shorter.

Conclusion

- Imitation gives human technique; RL on a KL leash adds wins, not spam.
- Next: all six stages, and games against people.

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